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Macroporosity of pasture topsoils after three years of set-stocked and rotational grazing by sheep

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Abstract. Grazing of livestock on pasture land can result in changes to the physical condition of soil, particularly as a result of trampling and changed organic matter status of the topsoil. Over time, changes to pasture botanical composition may also occur, which might further affect soil structure. The extent of the effects, and the rate of soil and pasture recovery when livestock are removed, will depend on the grazing management tactics employed. A field study was established at Orange, New South Wales, to compare soil physical properties under 'set-stocked' grazing of Merino sheep, 'high intensity–short duration' rotational grazing, an ungrazed control, and pasture cages. Topsoil bulk density, hydraulic conductivity, and organic carbon content were measured annually over 3 years, and image analysis of soil macroporosity was carried out annually to quantify changes in pore geometry. Only the topsoil macropore properties changed significantly between treatments over the 3 years. In particular, the structural quality of the topsoil under set-stocked grazing changed, as indicated by a decrease in total macroporosity and a smaller proportion of macropores. In contrast, stable structural conditions were maintained under rotational grazing. Possibly the best soil structure for plant growth, represented by high values of total macroporosity and macropore surface area, and a large range of pore sizes, was exhibited under the pasture cages, where pasture defoliation occurred in the absence of hoof pressure. It is concluded that grazing tactics are an important factor in the dynamics of soil macroporosity and the vertical continuity of macropores, as a result of the combined effects of hoof pressure and root channel development.

Introduction

The impacts of grazing livestock on extensive pasture production systems include soil compaction from hoof pressure, physical damage to pasture plants, and changes to pasture botanical composition over time (Kemp *et al*. 2000; Drewry *et al*. 2008). In response to the resultant decline in pasture productivity and a greater focus on environmental effects, improvements to grazing management strategies are under consideration in the main grazing areas of south-eastern Australia, with a view to better management of the perennial component of the pasture (Michalk *et al*. 2003).

The traditional grazing management strategy has been set-stocking, where a fixed number of livestock are held continuously in each paddock and allowed to consume palatable pasture species selectively (Latham and Murray 1996). The continuous presence of hoof pressure under such grazing inhibits regeneration of perennial pasture species and often results in changes to pasture botanical composition. These factors have been associated with declining pasture productivity and persistence, and suboptimal livestock performance (Kemp and Dowling 2000).

Various strategies of 'planned grazing' have received attention from graziers in an attempt to enhance the perennial species component of pastures. One such strategy is high intensity–short duration (HI–SD) grazing (Heitschmidt 1993), where large numbers of livestock are put onto pasture for a short time when conditions are satisfactory, followed by a rest period after removal of the stock. The grazing pressure, as measured by the number and class of livestock, the timing and duration of the grazing event, and the duration of the rest period, is determined by the manager according to the stage of growth of the pasture, ground cover before and after grazing, and the expected pasture growth during the rest period. One version of HI–SD grazing is 'cell' grazing, where larger grazing paddocks are divided into many smaller grazing cells, with the livestock rotated between cells according to a relatively strict grazing plan. A comparison of alternative grazing tactics and their likely impact on pasture productivity and botanical composition is provided by Fitzgerald and Lodge (1997) and Kemp *et al*. (2000).

Strategies that promote perennial pasture species are likely to increase dry matter production and pasture persistence, and may provide environmental benefits by increasing soil water use, reducing soil acidification, and reducing dependence on chemical control of weeds (Kemp *et al*. 2000). However, much of the recent research has focussed on the above-ground performance of such systems, since this controls the short-term economic performance of the enterprise, but HI–SD grazing strategies may also influence soil structure and, as a

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consequence, the long-term performance and sustainability of the enterprise.

Compaction pressure under livestock can be substantial (Packer 1988; Greenwood *et al*. 1997; Betteridge *et al*. 1999; Greenwood and McKenzie 2001), and this contributes directly to a reduction in topsoil porosity and vertical pore continuity (Greenwood and McKenzie 2001) through the disruption of aggregates into smaller particles and the repacking of smaller particles to fill existing voids. The process of compaction by livestock treading may be particularly important when the soil is wet (Proffitt *et al*. 1995*a*), although in the case of sheep treading, such compaction is usually restricted to the upper 50–150 mm of the soil profile (Greenwood and McKenzie 2001). Comparisons of physical properties of ungrazed and grazed topsoils have demonstrated significantly greater bulk densities (Witschi and Michalk 1979; Proffitt *et al*. 1995*a*; Dormaar and Willms 1998) and soil strengths (Proffitt *et al*. 1995*b*), and significantly lower hydraulic conductivities (Proffitt *et al*. 1993; Greenwood *et al*. 1997; Greenwood *et al*. 1998), under high stocking rates. However, it is not clear whether the post-grazing rest periods incorporated into HI–SD grazing strategies are sufficient to provide decompaction benefits associated with increased organic matter turnover and the removal of hoof traffic.

Relatively few studies have focussed on soil structure deterioration and subsequent recovery under grazing systems (Drewry 2006; Drewry *et al*. 2008). Greenwood *et al*. (1998) found that 2.5 years of grazing exclusion from degraded topsoil in northern New South Wales (NSW) led to significantly higher unsaturated hydraulic conductivity and lower bulk density than in topsoil subjected to continuous grazing. In New Zealand, Drewry *et al*. (2004) observed that dairy farm topsoil became denser during spring as a result of cattle treading, but then recovered (i.e. decrease in bulk density) over summer and autumn despite the continued rotational grazing of cows. This recovery was attributed to the natural drying and cracking processes of the soil. Similarly, in another New Zealand study, Nie *et al*. (1997) measured an increase in topsoil hydraulic conductivity and a small reduction in topsoil bulk density following 7 months of pasture rest ('pasture fallow') between spring and autumn. However, Greenwood and McKenzie (2001) cite little evidence to support the potential benefits of HI–SD grazing on soil physical condition, finding no published research in Australia to indicate a rapid recovery of soil structural condition following the removal of grazing pressure.

Perhaps the main reason for the lack of information on the effects of grazing management on soil structure deterioration and recovery is that the selection of appropriate indicators is problematic. Easy-to-use, rapid, indirect measures of soil structural condition, such as bulk density and penetration resistance, do not provide information on pore size distribution or pore connectivity, while most quantitative, direct measurements of porosity are time-consuming and require specialised analytical equipment. Although requiring destructive testing methods, image analysis of soil core sections, from either the horizontal or vertical plane, has proven useful in various studies of soil structure (e.g. Douglas *et al*. 1992; Koppi *et al*. 1992; McKenzie 2001). Image analysis enables direct measurement of pore size, shape, and surface area and, when multiple stacked sections are analysed, indirect assessment of pore connectedness. Knowledge of the distribution of pores throughout the topsoil allows assessment of the effects of compaction from livestock hoof pressure. The method can detect small differences in pore geometry and is therefore more sensitive to differences in soil management than indirect (surrogate) methods.

A 3-year grazing experiment using Merino sheep was established at Orange, NSW, to compare the effects of set-stocked grazing and HI–SD grazing on soil structural condition. This paper reports topsoil macropore attributes of the differently managed grazing areas, in addition to several associated topsoil properties, to quantify differences in topsoil structure arising from alternative grazing tactics.

Materials and methods

Experimental site

The experimental site is a 6-ha field at the former campus of The University of Sydney, Orange, NSW, centred at 310 900 m east and 1 319 250 m north, with an altitude of 875 m ASL. The field has a generally easterly aspect, with a relatively uniform 3% surface gradient and unimpeded surface drainage. A grassed, 6-m-wide buffer strip separated the site from adjoining grazed land. The site has a long history of grazing by sheep and cattle, occasional fertiliser application, and occasional machine operations associated with pasture establishment and fodder conservation. The experimental site was developed during 1999, with livestock introduced in December of that year. Because of dry seasonal conditions before this, the field was left ungrazed during 1997, and pasture was cut for hay during 1998. Therefore, livestock had not been in the field for more than 2 years before the experiment began. Prior to the introduction of livestock, the dominant species were perennial ryegrass (*Lolium perenne*), Yorkshire fog (*Holcus lanatus*), and subterranean clover (*Trifolium subterraneum*), with phalaris (*Phalaris aquatica*), white clover (*Trifolium repens*), vulpia (*Vulpia bromoides*), barley grass (*Hordeum glaucum*), annual ryegrass (*Lolium rigidum*), and Patterson's curse (*Echium plantagineum*) also present.

Climate

The climate at the experimental site is typical of the Central Tablelands of NSW, with cold, wet winters and mild summers. Rainfall was above average in 2 of the 3 months before the livestock were introduced, and annual rainfall was slightly above average during 2000. In the next 2 years, below-average annual rainfall was recorded at the site. Seasonal rainfall records, particularly for the critical spring/summer pasture growth periods, indicate below-average summer rainfall for 2001/02 and 2002/03, coinciding with above-average mean maximum and mean minimum temperatures for spring/summer 2002/03. Despite these trends, supplementary feeding of livestock was not required during the experiment.

Soil properties before the introduction of livestock

A 1.5-m-deep trench was excavated adjoining the experimental site. Here, the profile is a shallow, well-structured Brown Dermosol (Isbell 1996). Topsoil field textures are loamy, and become gradually more clayey with depth. Below ~1 m, highly weathered mudstone/siltstone dominates the profile, limiting deep drainage in wet conditions. Field observations showed a strongly pedal structure, with 30–50-mm sub-angular blocky peds primarily in the upper 200 mm, and 5–10-mm polyhedral peds throughout the profile. Abundant plant roots and frequent faunal passages were present in the upper 200 mm of the profile, decreasing with depth.

To establish benchmark values and to assess soil variation across the site, several topsoil attributes were measured across the site before livestock were introduced. Using a 25-m grid sampling scheme, 111 topsoil samples were taken from 0–0.1 m depth and analysed for bulk density (using 50 by 50 mm cylindrical steel cores), pH (1 : 5, soil : water extract), electrical conductivity (1 : 5, soil : water extract), and clay content (using hydrometer measurements). Surface hydraulic conductivity was also measured at 25 of these sampling locations. In addition, commercial soil tests conducted over the 3 years before the experiment provided data on exchangeable cation concentrations. The average values of these topsoil attributes are given in Table 1.

The higher (western) side of the field had slightly greater average clay contents and slightly smaller average bulk densities than the lower (eastern) side of the field. Principal component analysis revealed that the differences between the western and eastern sides were more apparent when multiple parameters were combined, and so to ensure no bias between treatments, equal numbers of plots were randomly allocated to both grazing treatments on both the western and eastern sides of the field (Fig. 1).

Grazing treatments

The site was divided into 0.8-ha plots, and subjected to 2 grazing treatments using Merino wethers and an ungrazed control. There were 4 replications of each treatment (Fig. 1) and treatments were imposed from 20 December 1999 to 12 February 2003 (Table 2). Merino wethers were selected for their uniform nutritional requirements over the seasons, and for ease of management. The plot size was regarded as sufficient to allow 'normal' grazing behaviour.

Treatment SS was initially set-stocked with 10 dry sheep equivalents (dse) per ha, which was increased to 12 dse/ha for Year 3. Treatment HI–SD was managed by high intensity, short duration grazing events (up to 220 dse/ha for 1–3.5 days), depending on pasture availability. The timing of these grazing events is given in Table 2. Livestock were removed if surface water was present. The control areas were maintained

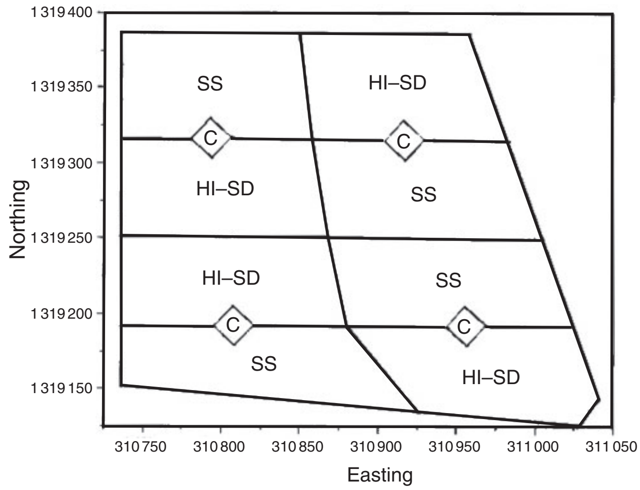


Fig. 1. Schematic layout of plots and treatments. Treatments: SS, set-stocked; HI–SD, high intensity short duration; C, control.

Table 1. Mean values of topsoil (0–0.1 m depth) properties before the introduction of livestock

Numbers in parentheses are standard errors of the means, and data for exchangeable cations are based on historical soil tests of bulked samples

Topsoil property	Mean value
pH	6.1 (0.035)
Electrical conductivity (dS/m)	0.047 (0.001)
Clay content (%)	17.3 (0.32)
Bulk density (g/cm^3)	1.29 (0.008)
K_{10} (mm/h)	26.7 (4.84)
Cation exchange capacity (cmol(+)/kg)	16.6
Exchangeable Na %	1.26
Exchangeable Ca/Mg ratio	1.86

Table 2. Chronological sequence of grazing and soil sampling events

Date range	Grazing treatment	Soil sampling/ measurement
	Year 0	
October 1998	–	pH, EC, bulk density, clay
March 1999		Hydraulic conductivity
	Year 1	
20 December 1999	SS commenced	–
February 2000 (2 days)	HI–SD grazing events	–
May 2000 (3.5 days)		
October 2000 (3 days)		
Nov./Dec. 2000 (3 days)		
January 2001 (2.5 days)		
27 January 2001		Hydraulic conductivity
9 February 2001		Bulk density, organic carbon
27 February 2001		Image analysis cores
	Year 2	
April 2001 (3 days)	HI–SD grazing events	–
October 2001 (2 days)		
December 2001 (2.5 days)		
13 January 2002		Bulk density, organic carbon, hydraulic conductivity
12 February 2002		Image analysis cores
	Year 3	
March 2002 (2 days)	HI–SD grazing events	–
September 2002 (3.5 days)		
December 2002 (3 days)		
18 January 2003		Image analysis cores
24 January 2003		Bulk density, organic carbon
6 February 2003		Hydraulic conductivity
12 February 2003	SS ceased	–

by stockproof exclosures (treatment C). Although not replicated at the plot scale as an additional treatment (but labelled treatment CA), wire cages were deployed at randomly selected locations within each grazed plot. Pasture plants growing through these cages were grazed from above, but livestock were prevented from treading the soil under the cages.

The total grazing load was approximately equal for each of the 2 grazing treatments when averaged over the entire experiment. In the case of treatment SS, this was 10.5 dse/ha, and in the case of treatment HI–SD, 9.1 dse/ha, the difference representing only 1 HI–SD grazing event. However, livestock were present on treatment HI–SD for only 30 days in total, mostly during spring and summer when pasture growth was substantial. Pasture composition of the differently managed plots was measured on an annual basis throughout the experiment, but as there were no significant differences in this attribute between treatment plots at any time, these data are not presented.

Soil sampling regime and measurements

Following the introduction of Merino wethers to the experimental plots in December 1999, topsoil attributes were measured on an approximately annual basis for 3 years. Because of the time needed to undertake all soil measurements, their sensitivity to soil moisture content, and the need to align with grazing events, not all samples and measurements could be done at the same time in each year, nor 12 months apart. The timing of soil sampling events and soil measurements is given in Table 2. To avoid the effects of stock walking tracks and camps, soil samples were not collected within 5 m of any fence.

Field methods for soil measurements

The bulk density of topsoils was estimated by extracting cylindrical steel cores (50 by 50 mm) of soil (6 samples per plot), oven-drying the cores, and then calculating the mass of dry solids in the volume of the cores. A CSIRO tension disc permeameter was used to estimate the unsaturated hydraulic conductivity (K_{10}) of the soil surface (6 measurements per plot), using the method described by Geering (1995). The water tension was set at 10 mm to exclude the possibility of bypass flow due to surface cracking or the random presence of other large, surface-connected macropores. Cylindrical cores of undisturbed topsoil, to be used for image analysis measurements, were excavated by hand using 160-mm-diameter PVC tube (soil depth 140 mm, 3 samples per plot for treatments SS and HI–SD, 8 samples for treatments C and CA).

Laboratory methods for soil measurements

The 160-mm-diameter soil cores taken from the field site were flooded with Araldite resin containing diluent and fluorescent dye, based on the method described by Moran *et al*. (1989). The use of diluent and the method of irrigation allowed the resin to displace water and air, and occupy the soil pores. Once set hard, the cores were sectioned to prepare horizontal surfaces at depths of 10, 50, and 100 mm; these depths span most of the topsoil zone thought to be potentially compacted by livestock treading (Greenwood and McKenzie 2001; Drewry 2006). Under ultraviolet light, soil pores were easily differentiated from solids due to the fluorescent dye in the resin, and binary digital images of each horizontal surface were taken over a 150-mm-diameter area to avoid edge effects. The resolution of the images allowed resin-filled pores of $\geq$0.065 mm diameter to be detected. Although 'macropores' are not consistently defined in the literature, Luxmoore *et al*. (1990) and Drewry *et al*. (2008) reviewed pore terminology and showed that many pore classification schemes use a lower pore diameter of 30–100 µm to define 'macropores'. Consequently, all pores detected in the images collected here will be regarded as 'macropores'.

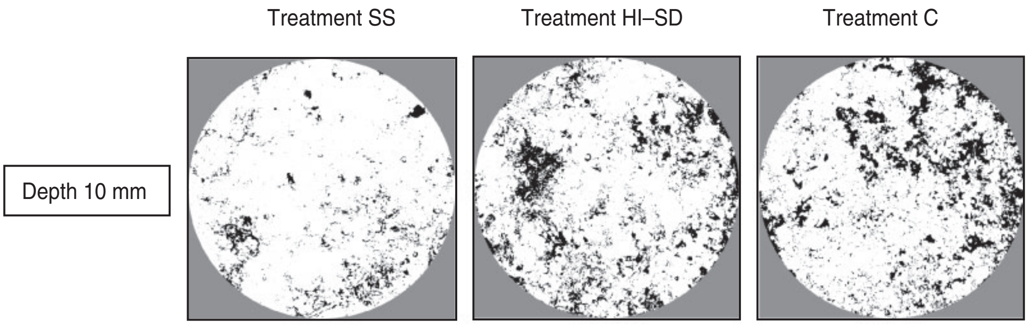
Each image was analysed using the SOLICON® computer program (Cattle *et al*. 2000) to provide data on macropore attributes at each depth, and the average of 3 topsoil depths. The following macropore attributes were measured: total macroporosity (proportion of pore space >0.065 mm diameter in the total soil volume, mm^3/mm^3); macropore surface area (mm^2/mm^3); macropore count density (number of macropores/mm^2); macropore shape (as determined by the degree of roundness of macropores, 1.0 for circular macropores); and macropore sieve (mean diameter of macropores if they were all spherical in shape, mm). To improve the correlation between unsaturated hydraulic conductivity and image analysis data, results for total macroporosity were also partitioned into 2 size ranges: macropores with diameter 0.065–1.5 mm ('macroporosity$_{<1.5}$'), and pores with diameter >1.5 mm ('macroporosity$_{>1.5}$'). This partitioning was carried out to separate macropores likely to be 'hydrologically active' (macropores >1.5 mm diameter; cf. Luxmoore *et al*. 1990) from those unlikely to be contributing significantly to water transport. Some sample images are shown in Fig. 2, illustrating the typical range of macropore attribute values obtained for the different treatments and sampling depths.

The total organic carbon contents of topsoil samples (6 samples per plot) were measured by the Walkley-Black method using the procedure described in Rayment and Higginson (1992).

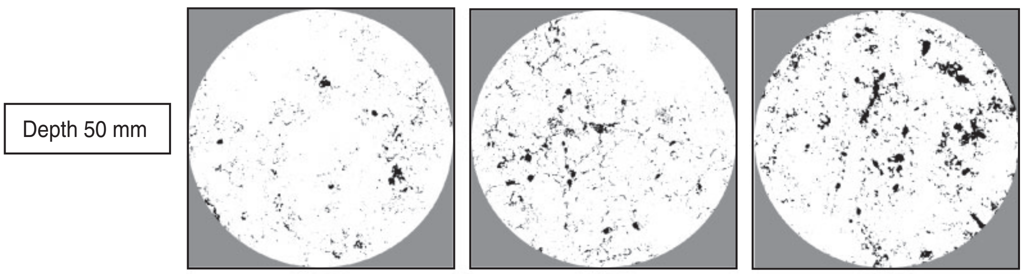
Statistical analysis

Hydraulic conductivity data were log-transformed before analysis, whereas all other data met the criteria for normality and equality of variance to a reasonable degree. Outliers were deleted from 2 image analysis datasets; these were assumed to be caused by sampling defects in the field and resin irrigation failure. Data were evaluated by analysis of variance using JMP *v*.3.2.2 software (SAS 1998). Differences between means were compared using the Tukey–Kramer h.s.d. technique at $P = 0.05$. Where correlations are shown, the Pearson correlation coefficient was estimated.

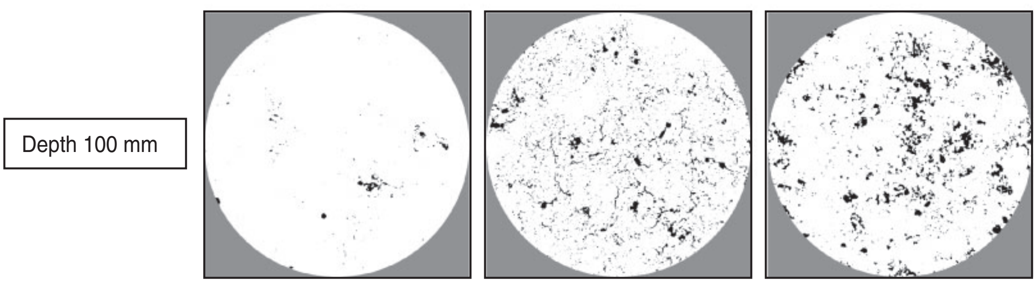
The relationships between soil properties were explored using linear correlation and principal component analysis. Principal component analysis was selected because of its potential usefulness in interpreting pedological meaning from multivariate datasets (Webster and Oliver 1990) according to the strength of alignment of soil attribute clusters with yet-to-be-determined factors, a method applied successfully by Brejda *et al*. (2000*a*, 2000*b*) in identifying regional soil quality indicators. The method finds the orthogonal axes of the multidimensional array of the dataset where correlation is



	Treatment SS	Treatment HI–SD	Treatment C
Total macroporosity (%)	6.58	17.4	18.8
Macropore surface area (mm^2/mm^3)	0.43	0.87	0.81
Macropore shape factor	0.53	0.46	0.47
Macropore count (no./mm^2)	0.12	0.18	0.18
Macropore sieve (mm)	0.44	0.56	0.71



	Treatment SS	Treatment HI–SD	Treatment C
Total macroporosity (%)	2.65	6.26	8.43
Macropore surface area (mm^2/mm^3)	0.19	0.33	0.38
Macropore shape factor	0.57	0.57	0.58
Macropore count (no./mm^2)	0.07	0.17	0.12
Macropore sieve (mm)	0.45	0.44	0.70



	Treatment SS	Treatment HI–SD	Treatment C
Total macroporosity (%)	0.60	4.70	7.06
Macropore surface area (mm^2/mm^3)	0.03	0.49	0.43
Macropore shape factor	0.57	0.58	0.59
Macropore count (no./mm^2)	0.01	0.11	0.09
Macropore sieve (mm)	0.54	0.37	0.55

Fig. 2. Sample images from 3 depths of 1 site in each of the 3 treatments, with corresponding SOLICON estimates of structural attributes. All sample images represent an actual diameter of 150 mm.

maximised. The principal components which represent the majority of the variance can be rotated in an attempt to redistribute the variance to maximise the relationship between interdependent attributes. The strength of alignment of an attribute with the rotated principal components, or factors, can be measured by the simple correlation between the attribute and the factor (the factor score). More importance can be placed on attributes with a high proportion of variance explained by the factor, measured by the communality of the attribute. While this type of factor analysis enables relationships between attributes to be explored, the factors have no prescribed physical meaning.

Results

Treatment effects on topsoil organic carbon, bulk density, and hydraulic conductivity

The topsoil organic carbon contents varied little between treatment plots or between the 3 years of measurement (Table 3). The only significant difference observed was that the organic carbon content of soil in treatment C was significantly smaller than for both grazing treatment soils at the conclusion of the experiment. The reason for this is unclear, but the pasture sward in treatment C plots became dominated by senescent *Phalaris aquatica*, reducing pasture species diversity and possibly vegetative matter production. In the grazed treatments, continuous or periodic defoliation may have contributed to accelerated organic matter accumulation through increased root regeneration and decay of root and surface material (Kemp *et al*. 2000). The difference is of little practical significance because the total organic carbon content is similar to that of the larger values measured under pasture in other experiments (Geeves *et al*. 1995) and is well above the generally suggested lower limit of 1% for maintenance of soil structure (Charman and Roper 2000).

There were no significant differences in topsoil bulk density between treatments in any year (Table 3). Mean bulk densities measured in this experiment were generally at the lower end of the range of values summarised by Greenwood and McKenzie (2001) from many experiments and various soil types carrying pastures grazed by sheep.

Differences in unsaturated hydraulic conductivities of topsoils (K_{10} values) between treatments were not consistent across all years (Table 3). Some significant differences in log K_{10} values were observed between treatments in the first 2 years of the experiment, with topsoil of treatment SS exhibiting significantly smaller log K_{10} values than topsoil of treatment CA. At the end of the experiment, however, no significant differences were observed in log K_{10} values between treatments, and no apparent trend with time was observed. All values were within the range of values for K_{10} reported by Lawrie *et al*. (2000) from grazed pastures on a large number of experimental sites across south-eastern Australia.

Treatment effects on soil structural attributes

Total macroporosity

After 3 years of alternative grazing management, the average total macroporosity of topsoil under treatment SS was significantly less than that of topsoil from treatments HI–SD and CA, with this difference becoming established in Year 2 (Table 4). Under treatment SS, the average total macroporosity showed a continuous reduction, with average values being significantly less in Year 3 than Year 1, while average total macroporosity for treatments HI–SD and CA remained the same or increased. The largest values of average total macroporosity in Years 2 and 3 were consistently measured where pasture was grazed through cages, which prevented hoof pressure on the soil surface. There was no consistent trend in average total macroporosity values for soil subjected to treatment C, although average values in Years 2 and 3 were significantly less than that of Year 1.

By Year 2, the soil of treatment SS tended to have smaller total macroporosity than the soils of all other treatments at each depth; this difference was significant compared with treatment CA. At the end of the experiment, differences in total macroporosity between soil of treatments SS and HI–SD were significant at all sampling depths. Total macroporosity under treatment HI–SD was similar to that under treatment CA at all depths. For all treatments, total macroporosity at a depth of 10 mm was consistently 2–3 times greater than at depths of 50 and 100 mm (Table 4, Fig. 2). Total macroporosity at 10 mm depth declined appreciably after Year 1 in all treatments except treatment CA.

Macroporosity$_{<1.5}$

The trends for values of the average macroporosity <1.5 mm diameter are similar to the trends for average total macroporosity. After Year 2, the topsoil of treatment SS had

Table 3. Average values of topsoil attributes for plots of different grazing treatments
Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Attribute	SS	HI–SD	C	CA
		Year 1		
Bulk density (g/cm^3)	1.08 (0.03)	1.03 (0.03)	1.12 (0.04)	1.01 (0.05)
Organic C (%)	2.6 (0.1)	2.4 (0.1)	2.4 (0.2)	
Log K_{10} (log mm/h)	1.24 (0.03)a	1.28 (0.05)ab	1.41 (0.06)ab	1.48 (0.10)b
K_{10} (mm/h)	18.7	21.3	27.8	35.5
		Year 2		
Bulk density (g/cm^3)	1.14 (0.02)	1.16 (0.03)	1.20 (0.03)	1.15 (0.08)
Organic C (%)	2.6 (0.1)	2.8 (0.1)	2.3 (0.2)	
Log K_{10} (log mm/h)	1.17 (0.04)a	1.34 (0.05)b	1.28 (0.07)ab	1.41 (0.09)b
K_{10} (mm/h)	14.8	21.9	19.0	25.9
		Year 3		
Bulk density (g/cm^3)	1.05 (0.03)	0.99 (0.03)	1.11 (0.03)	1.05 (0.04)
Organic C (%)	2.8 (0.1)a	2.9 (0.1)a	2.5 (0.1)b	
Log K_{10} (log mm/h)	1.37 (0.06)	1.39 (0.04)	1.35 (0.19)	1.42 (0.07)
K_{10} (mm/h)	27.9	26.5	26.9	28.9

Table 4. Total macroporosity values (%) at 3 depths, and depth-averaged values of total macroporosity, macroporosity <1.5 mm in diameter, and macroporosity >1.5 mm in diameter, for different grazing treatments
Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Property	SS	HI–SD	C	CA
		Year 1		
Total at 10 mm depth	21.5 (2.6)	22.2 (3.0)	30.2 (3.0)	26.3 (4.1)
Total at 50 mm depth	6.0 (1.0)a	6.9 (0.7)ab	10.2 (01.6)b	8.7 (1.3)b
Total at 100 mm depth	3.7 (0.6)a	4.1 (0.6)a	7.9 (2.2)b	6.8 (1.1)b
Average total	10.4 (1.2)	11.0 (1.2)	16.1 (0.8)	13.9 (2.0)
Average <1.5 mm	10.1 (1.2)	10.4 (1.1)	14.4 (1.0)	10.7 (1.2)
Average >1.5 mm	0.35 (0.09)a	0.66 (0.13)a	1.65 (0.45)ab	3.20 (0.84)b
		Year 2		
Total at 10 mm depth	14.3 (0.7)a	17.2 (1.2)ab	15.1 (2.2)ab	22.5 (1.9)b
Total at 50 mm depth	4.7 (0.4)a	5.8 (0.8)ab	6.5 (0.8)ab	8.7 (1.1)b
Total at 100 mm depth	2.9 (0.3)a	4.2 (0.6)ab	4.7 (0.9)ab	7.0 (1.0)b
Average total	7.3 (0.3)a	9.8 (0.5)ab	8.7 (1.0)ab	12.0 (1.0)b
Average <1.5 mm	6.9 (0.3)a	8.9 (0.7)ab	7.1 (0.6)ab	10.2 (0.7)b
Average >1.5 mm	0.37 (0.05)a	0.86 (0.15)ab	1.61 (0.38)bc	1.98 (0.34)c
		Year 3		
Total at 10 mm depth	9.3 (1.2)a	18.1 (2.1)b	15.9 (1.6)ab	22.7 (2.4)b
Total at 50 mm depth	4.4 (0.7)a	7.6 (1.2)b	7.5 (0.8)ab	8.0 (0.10)b
Total at 100 mm depth	3.8 (0.6)a	8.0 (0.7)b	4.7 (0.3)ab	7.5 (1.1)b
Average total	5.6 (0.7)a	12.4 (0.7)b	9.4 (0.8)ab	12.7 (0.9)b
Average <1.5 mm	4.6 (0.7)a	10.3 (0.6)b	8.0 (0.7)ab	10.0 (0.7)b
Average >1.5 mm	0.88 (0.17)a	2.09 (1.60)a	1.43 (0.02)ab	2.67 (0.41)b

significantly smaller average macroporosity$_{<1.5}$ than the soil under the CA treatment, with this trend continuing to the end of the experiment (Table 4). Average macroporosity$_{<1.5}$ was greatest in Year 1 for all treatments, remaining stable throughout the experiment in treatments HI–SD and CA.

Macroporosity$_{>1.5}$

Although the average macroporosity >1.5 mm diameter was a relatively small proportion of average total macroporosity (Table 4), there were interesting differences between treatments. In all 3 years, soil of both grazing treatments had a significantly smaller proportion of macroporosity$_{>1.5}$ than soil of treatment CA, when averaged over the full 100 mm depth of sampling (Table 4). Only in Year 2 did treatment C show significantly greater average macroporosity$_{>1.5}$ than treatment SS. Soils of the HI–SD and C treatments did not differ significantly for average macroporosity$_{>1.5}$ in any year. For both grazing treatments, the maximum contribution of average macroporosity$_{>1.5}$ to average total macroporosity was observed in Year 3, which was not the case for the ungrazed treatments.

Macropore surface area

At the end of Year 1 there were no significant differences in topsoil macropore surface area between any of the treatments (Table 5). In Year 2, significant differences in macropore surface area emerged at certain depths for some treatments, and by Year 3 average macropore surface area was significantly smaller for soil in treatment SS than in HI–SD and CA. The smaller macropore surface area under treatment SS occurred at each depth of measurement, at the end of Year 3, a trend that mirrors results for total macroporosity. Sample images are presented in Fig. 3 and show the variation in macropore surface area; images dominated by large, cylindrical macropores tend be soils that have small values of macropore surface area.

Macropore count density

Significant differences in macropore count density were observed between soils of treatments HI–SD and C in Year 2, and between soils of treatments SS and HI–SD in Year 3, with fewer macropores in treatment SS than HI–SD (Table 6). A declining trend in average macropore count with time was observed for all treatments. A generally declining trend for macropore count was also observed with depth, most apparent for treatment SS. Sample images illustrate the variation in macropore count density (Fig. 3).

Macropore shape factor

Some significant differences in macropore shape were observed at Year 2, with a more rounded average macropore shape factor in soil of treatment HI–SD compared with treatment CA (Table 7); however, the differences in absolute terms appear small. A generally increasing (more rounded) trend in average macropore shape factor was observed with time. Treatment SS showed a significantly larger macropore shape factor at the soil surface for Years 2 and 3, and there was a generally increasing macropore shape factor with depth. Sample images illustrate the variation in macropore shape factor (Fig. 4).

Macropore sieve

Soil of treatment CA had significantly greater average macropore sieve values than soil of both grazing treatments

Table 5. Macropore surface area (mm^2/mm^3) at each depth of measurement for different grazing treatments

Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Depth (mm)	SS	HI–SD	C	CA
		Year 1		
10	1.05 (0.12)	1.05 (0.12)	1.09 (0.04)	0.72 (0.03)
50	0.42 (0.07)	0.50 (0.06)	0.67 (0.09)	0.42 (0.07)
100	0.28 (0.05)	0.31 (0.06)	0.49 (0.14)	0.28 (0.03)
Average	0.59 (0.07)	0.62 (0.06)	0.75 (0.09)	0.47 (0.04)
		Year 2		
10	0.69 (0.03)ab	0.71 (0.04)ab	0.50 (0.06)a	0.74 (0.038)b
50	0.30 (0.03)	0.35 (0.04)	0.29 (0.03)	0.45 (0.03)
100	0.21 (0.02)a	0.25 (0.04)a	0.21 (0.03)a	0.38 (0.03)b
Average	0.40 (0.02)ab	0.48 (0.03)ab	0.33 (0.03)a	0.52 (0.03)b
		Year 3		
10	0.40 (0.05)a	0.60 (0.05)b	0.48 (0.05)ab	0.62 (0.04)b
50	0.18 (0.03)a	0.33 (0.06)ab	0.35 (0.05)ab	0.33 (0.05)b
100	0.15 (0.03)a	0.34 (0.04)b	0.22 (0.02)ab	0.31 (0.06)ab
Average	0.23 (0.04)a	0.45 (0.04)b	0.35 (0.05)ab	0.42 (0.04)b

for Years 1 and 2 (Table 8). Greater average macropore sieve was measured for treatments SS, HI–SD, and CA in Year 3 than Years 1 and 2. There were no significant differences found between treatments at the conclusion of the experiment.

Soil of treatment SS had significantly smaller macropore sieve values than soil of treatment CA at most depths in Years 1 and 2. In Year 2, significant differences in macropore sieve values between treatment SS and all other treatments were observed at various depths. By Year 3, the difference between treatment SS and treatment CA was significant only at 10 mm depth; at 100 mm depth the macropore sieve values were similar for all treatments. Sample images are presented in Fig. 4 and illustrate variation in macropore sieve.

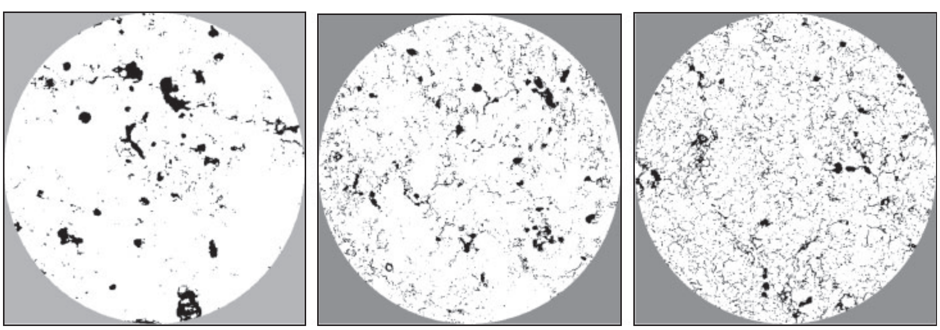
Relationships between soil properties

Pearson product moment correlation coefficients within each year are shown for selected soil attributes (Table 9). Total macroporosity at 100 mm depth was selected because of its potential as an indicator of macropore continuity. Correlation between image analysis results for average total macroporosity and average macroporosity$_{<1.5}$ was generally high, as expected, given the algorithms used to estimate these variables and their co-dependency. Consequently, the correlation between these properties and the more traditional surrogate measures of soil structure (bulk density and unsaturated hydraulic conductivity) is of most interest.

Correlation was seen between some of the image analysis results and the other measures of soil structure, most notably between macroporosity measurements and unsaturated hydraulic conductivity in Year 2, but only weak correlation was exhibited between macropore count, shape, or sieve and bulk density or unsaturated hydraulic conductivity. Strong correlation between macroporosity measurements and macropore count and macropore surface area is also apparent, less for macropore shape factor, and least for macropore sieve. There was a significant negative correlation between macropore shape factor and total macroporosity, implying that more rounded pores are associated with smaller values of macroporosity. The negative correlations between macropore sieve, surface area, and count are of interest; a higher macropore population appears to be associated with smaller macropores while providing a larger macropore surface area.

Principal component analysis

Principal component analysis revealed that the first 3 components accounted for 86.9% of the variance in Year 2 and 83.1% in Year 3, with data derived from image analysis contributing strongly to the first component in both cases. In Year 2, there was strong alignment of total macroporosity



(*a*) (*b*) (*c*)

	(*a*)	(*b*)	(*c*)
Total macroporosity (%)	7.13	7.09	9.71
Macropore surface area (mm^2/mm^3)	0.18	0.53	0.82
Macropore count density ($no./mm^2$)	0.04	0.17	0.25
Macropore shape factor	0.57	0.58	0.54
Macropore sieve (mm)	1.45	0.47	0.34

Fig. 3. Sample images from soil of treatments (*a*) SS, (*b*) CA, and (*c*) CA, illustrating variation in macropore surface area, macropore count density, and macropore sieve. All samples are from 50 mm depth and represent an actual diameter of 150 mm.

Table 6. Macropore count density (no./mm^2) at each depth of measurement for different grazing treatments
Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Depth (mm)	SS	HI–SD	C	CA
		Year 1		
10	0.165 (0.013)	0.175 (0.014)	0.136 (0.018)	0.127 (0.013)
50	0.124 (0.020)	0.143 (0.016)	0.175 (0.027)	0.093 (0.015)
100	0.083 (0.013)	0.103 (0.021)	0.131 (0.034)	0.103 (0.046)
Average	0.124 (0.013)	0.140 (0.012)	0.148 (0.025)	0.108 (0.021)
		Year 2		
10	0.142 (0.007)a	0.130 (0.007)a	0.079 (0.007)b	0.117 (0.005)ab
50	0.097 (0.008)ab	0.104 (0.010)ab	0.071 (0.006)a	0.119 (0.007)b
100	0.078 (0.010)	0.086 (0.014)	0.060 (0.007)	0.094 (0.010)
Average	0.106 (0.007)ab	0.110 (0.008)a	0.070 (0.004)b	0.110 (0.016)ab
		Year 3		
10	0.079 (0.010)	0.090 (0.007)	0.070 (0.007)	0.076 (0.008)
50	0.045 (0.007)	0.073 (0.012)	0.079 (0.013)	0.071 (0.010)
100	0.038 (0.007)a	0.080 (0.009)b	0.052 (0.010)ab	0.069 (0.013)b
Average	0.050 (0.007)a	0.083 (0.010)b	0.066 (0.011)ab	0.072 (0.008)ab

Table 7. Macropore shape factor at each depth of measurement for different grazing treatments
Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

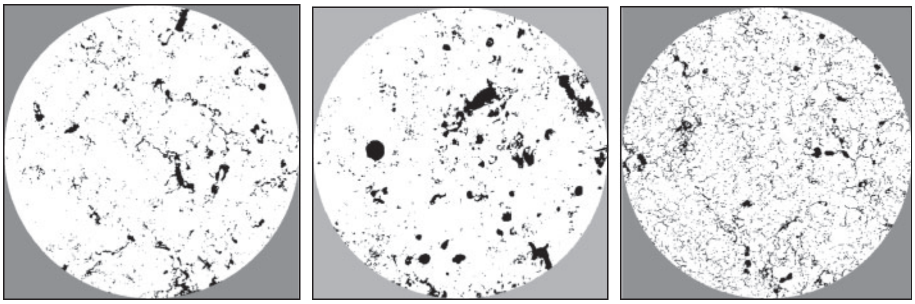
Depth (mm)	SS	HI–SD	C	CA
		Year 1		
10	0.449 (0.014)	0.448 (0.012)	0.407 (0.022)	0.449 (0.027)
50	0.553 (0.011)	0.521 (0.021)	0.484 (0.067)	0.572 (0.012)
100	0.556 (0.010)	0.519 (0.040)	0.565 (0.019)	0.578 (0.014)
Average	0.519 (0.010)	0.496 (0.021)	0.485 (0.025)	0.533 (0.014)
		Year 2		
10	0.505 (0.007)a	0.486 (0.007)ab	0.499 (0.025)ab	0.455 (0.010)b
50	0.578 (0.009)	0.588 (0.008)	0.570 (0.015)	0.555 (0.008)
100	0.580 (0.010)	0.645 (0.026)	0.616 (0.024)	0.568 (0.015)
Average	0.554 (0.006)ab	0.561 (0.008)a	0.562 (0.013)ab	0.526 (0.009)b
		Year 3		
10	0.529 (0.014)a	0.473 (0.015)b	0.486 (0.029)ab	0.495 (0.020)ab
50	0.607 (0.014)	0.575 (0.010)	0.580 (0.005)	0.563 (0.010)
100	0.602 (0.025)	0.643 (0.058)	0.609 (0.014)	0.611 (0.019)
Average	0.584 (0.014)	0.563 (0.021)	0.562 (0.011)	0.556 (0.008)

at 100 mm depth, average total macroporosity, average macroporosity$_{<1.5}$, macropore surface area, and macropore shape with Factor 1 (rotated principal component 1), although the communality for macropore shape was distinctly lower than that of the other attributes for Factor 1 (Table 10). Macropore count and macropore sieve aligned most strongly with Factor 2, and bulk density and unsaturated hydraulic conductivity aligned most strongly with Factor 3 (although the communality for unsaturated hydraulic conductivity was also low). For Year 3, a similar pattern of factor alignment emerged, although macropore surface area was more strongly aligned with Factor 2, and the low communality of macropore shape indicates weak alignment with any of the 3 factors in this model (Table 10).

The mean factor scores for each treatment in Years 2 and 3 are presented in Table 11. In Year 2, treatment CA was significantly different from all other treatments for Factor 1 and treatment SS was significantly different from treatment C for Factor 2. In Year 3, treatment SS is significantly different from treatments HI–SD and CA in Factor 1. There were no significant differences between treatments for Factor 2 or Factor 3 in Year 3.

Discussion

Examining only those soil attributes indirectly linked to soil structural condition, there appears little difference in the effects of alternative grazing tactics on soil physical condition. Similarly, there appear to be few effects of either grazing



(*a*) (*b*) (*c*)

	(*a*)	(*b*)	(*c*)
Total macroporosity (%)	9.83	8.59	9.71
Macropore surface area (mm²/mm³)	0.75	0.27	0.82
Macropore count density (no./mm²)	0.19	0.05	0.25
Macropore shape factor	0.30	0.61	0.54
Macropore sieve	0.34	1.25	0.34

Fig. 4. Sample images from soil of treatments (*a*) HI–SD, (*b*) C, and (*c*) CA, illustrating variation in macropore shape factor and macropore sieve with similar values of total macroporosity. All samples are from 50 mm depth and represent an actual diameter of 150 mm.

Table 8. Macropore sieve (mm) at each depth of measurement for different grazing treatments

Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Depth (mm)	SS	HI–SD	C	CA
		Year 1		
10	0.51 (0.03)a	0.57 (0.02)a	0.73 (0.09)a	1.18 (0.19)b
50	0.46 (0.04)a	0.42 (0.02)a	0.50 (0.07)ab	0.67 (0.08)b
100	0.40 (0.05)a	0.59 (0.14)ab	0.56 (0.08)ab	0.81 (0.08)b
Average	0.46 (0.03)a	0.53 (0.05)a	0.60 (0.05)ab	0.89 (0.09)b
		Year 2		
10	0.58 (0.03)a	0.67 (0.02)ab	0.94 (0.21)bc	0.93 (0.07)bc
50	0.49 (0.03)a	0.60 (0.06)ab	0.81 (0.06)b	0.70 (0.06)ab
100	0.50 (0.04)a	0.92 (0.18)b	0.81 (0.10)ab	0.66 (0.07)ab
Average	0.52 (0.03)a	0.64 (0.03)b	0.85 (0.08)c	0.76 (0.04)bc
		Year 3		
10	0.70 (0.06)a	0.85 (0.11)ab	0.95 (0.04)ab	1.14 (0.15)b
50	0.96 (0.17)	0.80 (0.05)	0.68 (0.08)	0.86 (0.08)
100	1.07 (0.15)	0.89 (0.10)	0.75 (0.15)	1.05 (0.11)
Average	0.92 (0.07)	0.89 (0.09)	0.80 (0.06)	1.02 (0.08)

regime on soil physical condition relative to the ungrazed control. One soil property expected to be sensitive to the grazing regime, total organic carbon content, did not significantly change under any of the treatments over the life of the experiment. This may be due to the relatively high and stable organic carbon content in all plots, the similarity of treatments in their impact on soil carbon, and the inability of the Walkley-Black method to separate and compare the labile soil carbon fractions. In contrast, Brejda *et al*. (2000*a*, 2000*b*) found that total organic carbon content was the single most useful indicator in discriminating between soil management practices in a study to identify those properties most useful as regional indicators of soil quality, although they also acknowledge that over short time intervals, more labile fractions of soil carbon may need to be evaluated.

Greenwood and McKenzie (2001) concluded that soil bulk density can be a surprisingly good indicator of soil change due to grazing; however, in this study no significant differences in soil bulk density were measured between treatments in any of the 3 years, a finding consistent with other research on grazing management and topsoil condition (e.g. Willatt and Pullar 1984; Abdel-Magid *et al*. 1987; Proffitt *et al*. 1993). It is likely that bulk density is a useful discriminator when grazing treatments have strongly contrasting effects, but an ineffective discriminator of subtle differences in grazing treatment effects.

Differences in unsaturated hydraulic conductivity between the plots of different treatments were few and inconsistent; only in Year 2 did the SS grazing plots exhibit significantly smaller unsaturated hydraulic conductivity than both the HI–SD rotational grazing and grazing over soil cages. Despite some studies demonstrating that topsoil hydraulic conductivity is significantly changed by grazing treatment (e.g. Greenwood *et al*. 1998), inconsistent changes in soil hydraulic properties have been a feature of other grazing experiments and reviews (e.g. Gifford and Hawkins 1978; Willatt and Pullar 1984; Abdel-Magid *et al*. 1987; Packer 1988; Proffitt *et al*. 1993; Rodd *et al*. 1999). This inconsistency in measured soil hydraulic properties is likely to be due to several factors, including the non-uniform distribution of macropores associated with the crowns and growing points of pasture plants (Greenwood *et al*. 1997), variation in the types of pasture plants present (Ridley 1996), and the duration of pasture growth before hydraulic conductivity measurements (Francis and Kemp 1990).

When examining the soil structural data obtained by image analysis and the results of the principal component analysis, a different set of effects of the grazing treatments becomes apparent. Average total macroporosity was shown to be generally increasing or stable with treatment HI–SD, and generally decreasing with treatment SS, with the differences significant after 3 years of grazing. Furthermore, the differences

Table 9. Pearson product moment correlation coefficients for selected soil attributes
MP, Macroporosity. $*P \leq 0.05$

	Log K_{10}	Bulk density	Total MP At 100 mm	Total MP Av.	Av. MP <1.5 mm	Av. MP >1.5 mm	Macropore Count	Macropore Surf. area	Macropore Shape
Year 1									
Bulk density	–0.16								
Total MP at 100 mm	0.12	–0.05							
Av. total MP	0.33	–0.35*	0.65*						
Av. MP <1.5 mm	0.33	–0.33	0.59*	0.95*					
Av. MP >1.5 mm	0.18	–0.23	0.50*	0.65*	0.38*				
Macropore count	0.18	–0.20	0.29	0.46*	0.58*	–0.05			
Surface area	0.25	–0.20	0.39*	0.68*	0.85*	–0.05	0.84*		
Shape	–0.06	0.12	–0.39*	–0.56*	–0.62*	–0.14	–0.47*	–0.62*	
Sieve	0.02	–0.21	0.36*	0.31	0.04	0.79*	–0.40*	–0.40*	0.16
Year 2									
Bulk density	–0.29								
Total MP at 100 mm	0.39*	0.29							
Av. total MP	0.40*	0.13	0.80*						
Av. MP <1.5 mm	0.43*	0.02	0.74*	0.98*					
Av. MP >1.5 mm	0.42*	0.06	0.76*	0.86*	0.73*				
Macropore count	0.16	–0.12	0.12	0.30	0.48*	–0.18			
Surface area	0.37*	0.04	0.60*	0.81*	0.90*	0.41*	0.80*		
Shape	–0.14	–0.22	–0.67*	–0.67*	–0.69*	–0.46*	–0.49*	–0.70*	
Sieve	0.18	0.01	0.35	0.33	0.17	0.72*	–0.58*	–0.20	0.19
Year 3									
Bulk density	–0.06								
Total MP at 100 mm	0.32	0.10							
Av. total MP	0.09	–0.10	0.66*						
Av. MP <1.5 mm	0.15	0.00	0.75*	0.96*					
Av. MP >1.5 mm	–0.09	–0.34	0.18	0.72*	0.51*				
Macropore count	0.24	0.11	0.74*	0.57*	0.74*	–0.08			
Surface area	0.20	0.05	0.80*	0.84*	0.94*	0.27	0.91*		
Shape	–0.14	–0.17	–0.51*	–0.43*	–0.56*	0.05	–0.64*	–0.63*	
Sieve	–0.33	–0.13	–0.37	–0.02	–0.23	0.51*	–0.66*	–0.43	0.52*

were maintained to at least 100 mm from the soil surface, indicating greater vertical continuity of macropores under HI–SD grazing and greater disruption of macropore continuity under SS grazing. At the end of the experiment, average total macroporosity and total macroporosity at each depth of measurement (10, 50, 100 mm) under HI–SD were similar to the maximum total macroporosity recorded, which was under the CA treatment. Similar trends were observed for macroporosity$_{<1.5}$ and for macroporosity$_{>1.5}$, providing evidence that differences in macropore density exist across a range of macropore sizes. While large pores have a critical influence over soil hydraulic properties, smaller pores are necessary to provide accessible, habitable, and protective pore spaces for microorganisms and the biotic interactions this facilitates (van Veen and Heijnen 1994; Shestak and Busse 2005).

These results are consistent with the findings of many other studies that have demonstrated a diminution of pore space with prolonged grazing pressure (e.g. Greenwood and McKenzie 2001; Drewry *et al.* 2008), but the range of macropore attributes measured in this study reveals more detail about how the macropore population has changed under the different grazing treatments. While the soil under treatment HI–SD showed a stable average total macroporosity over 3 years, and soil under treatment SS a decrease, macropore count declined for both (more so for treatment SS) and macropore sieve increased for both. Macropore shape became slightly more rounded for both treatments, but macropore surface area declined for both (more so for treatment SS). This indicates a subtle shift in the nature of the macroporosity over time, with the number of macropores in the topsoil reducing but the average macropore size slightly increasing. Such a shift can be observed by comparing Fig. 3*b* with Fig. 3*a*. This subtle change in topsoil structural form appears to be consistent with the destruction of finer macropores through compaction and the maintenance and/or establishment of macropores along plant root channels or as a result of soil macrofauna activity. These alterations to soil structure are apparent for topsoil of both grazing treatments, but more so for treatment SS, with soil of treatment HI–SD retaining a greater number of larger macropores. Since both treatments contained similar soil types, pasture species, and class of livestock, it is reasonable to attribute these differences to grazing method.

When compared with each other and with the data of the grazing treatment topsoils, the macropore attribute data of the C and CA treatment topsoils suggest some important distinctions between the effects of treading and defoliation during grazing.

Table 10. Rotated factor scores and communalities for soil attributes using a 3-factor model for Years 2 and 3 data

MP, Macroporosity

Soil attribute	Rotated factor scores			Communalities
Factor:	1	2	3	
		Year 2		
Bulk density	0.196	0.185	–0.891	0.868
Log K_{10}	0.324	0.255	0.272	0.699
MP at 100 mm	0.867	0.274	–0.092	0.836
Av. total MP	0.953	0.160	0.150	0.957
Av. MP < 1.5 mm	0.956	–0.033	0.136	0.934
Av. MP > 1.5 mm	0.752	0.593	0.153	0.941
Macropore				
Surface area	0.862	–0.442	0.140	0.959
Count	0.357	–0.876	0.158	0.919
Shape	–0.814	0.263	0.191	0.769
Sieve	0.204	0.896	0.126	0.860
		Year 3		
BD	0.122	0.388	0.717	0.680
Log K_{10}	0.203	0.344	–0.725	0.686
MP at 100 mm	0.839	0.066	–0.126	0.724
Av. total MP	0.854	–0.496	–0.075	0.980
Av. MP < 1.5 mm	0.947	–0.268	–0.041	0.970
Av. MP > 1.5 mm	0.320	–0.895	–0.134	0.921
Macropore				
Surface area	0.980	–0.019	–0.056	0.964
Count	0.877	0.335	–0.061	0.886
Shape	0.703	–0.313	–0.120	0.606
Sieve	–0.420	–0.784	0.167	0.818

Most of the significant differences in structural attributes between trodden (SS, HI–SD) and untrodden (C, CA) topsoils were between treatments CA and SS (Tables 4–8). There were some significant differences between treatment CA and treatment HI–SD topsoil, and relatively few significant differences between treatment C and either of the grazing treatment topsoils (Tables 4–8). These results confirm that while treading, and in particular the duration of treading, is an important determinant of compaction and topsoil macropore loss, moderate defoliation can be an important process for macropore renewal. In the ungrazed control plots, pasture composition quickly became dominated by mature phalaris, whereas the caged areas contained continuously active mixed pasture species of around 75 mm in height. It is likely that this resulted in different root system dynamics, and consequently differences in organic matter cycling, soil biota activity, and macropore maintenance/establishment. Tisdall and Oades (1980) measured differences in aggregate stability of a red-brown earth (Red Chromosol) when the ryegrass grown in it was subjected to different levels of defoliation by clipping (to simulate the effects of grazing without hoof pressure). Aggregate stability was greatest when the ryegrass was top-clipped at monthly intervals, associated with the presence of arbuscular mycorrhizal (fungal) hyphae. It was hypothesised that the clipping led to cycles of root decay and regeneration, stimulating hyphae production and leading to the additional release of water-soluble root exudates. Clipping at 2-week intervals was not as beneficial, probably because the time interval was insufficient to allow root recovery from the stress of clipping. This observation is consistent with the intention and practice of HI–SD grazing management.

The utility of the various soil macropore attributes for discriminating between grazing systems can be further assessed from the principal component analyses. For the data from both Year 2 and Year 3, the attributes of total macroporosity, macroporosity$_{<1.5}$, total macroporosity at 100 mm depth, macropore surface area, and macropore shape were most strongly aligned with the first principal component (Factor 1), indicating that these properties are the best discriminators of treatment effects. Of the macropore attributes aligned to Factor 1, macropore shape has the lowest communality (Table 10), indicating that this attribute accounts for comparatively less of the variance explained by the factor. This is likely to be due to the small differences in macropore shape between soil treatments, with the average macropore shape factor in Year 3 ranging from 0.562 to 0.585 across all treatments, and from 0.485 to 0.585 across the full dataset. Since the maximum possible range of macropore shape factor is 0–1, this indicates a very small variation in this property, which, as shown in Figs 3 and 4, is not as visually apparent as other macropore characteristics.

By considering multiple soil attributes simultaneously, this technique of data analysis provides clearer evidence of differences between treatments than examining each attribute separately. As shown in Table 11, in Year 2, the treatments SS and CA are significantly different from each other for Factor 1

Table 11. Mean factor scores for each treatment derived from soil data of Years 2 and 3

Standard error of the mean shown in parentheses, and in any row, pairs of values with the same letter are not significantly different at $P = 0.05$ (Tukey–Kramer test of honestly significant difference)

Factor	SS	HI–SD	C	CA
		Year 2		
1	–0.543 (0.134)a	–0.133 (0.240)a	–0.319 (0.362)a	1.228 (0.429)b
2	–0.611 (0.230)a	0.101 (0.463)ab	1.231 (0.199)b	0.229 (0.251)ab
3	–0.214 (0.156)	0.268 (0.341)	–0.225 (0.256)	0.265 (0.670)
		Year 3		
1	–0.855 (0.269)a	0.740 (0.281)b	–0.089 (0.271)ab	0.549 (0.264)b
2	0.488 (0.242)	–0.237 (0.528)	0.448 (0.076)	–0.686 (0.335)
3	0.045 (0.421)	–0.232 (0.306)	0.652 (0.440)	–0.017 (0.286)

scores, while by Year 3, treatments HI–SD and CA are both significantly different from treatment SS for Factor 1 scores. The lack of differences between treatments for Factor 3 scores, to which bulk density and hydraulic conductivity are aligned, suggests the superiority of image analysis over surrogate measures of soil structure to distinguish between treatments.

The implication of the significant differences in macropore properties of topsoil subjected to SS and HI–SD grazing treatments is that the periods of soil rest from grazing in the HI–SD strategy are effective in slowing structural change over time. While the recovery of soil structure from compaction events will be assisted by active pasture growth during periods of rest from grazing, the time needed to restore soil physical condition is likely to be highly variable, from some months to many years (Greenwood and McKenzie 2001; Drewry 2006). This variability is to be expected given the complex factors that determine pasture growth rates, including rainfall. However, to satisfy the needs of an intensive rotational grazing management system, some certainty is required over soil restoration patterns, preferably on a seasonal or annual basis to match seasonal management decisions on pasture management and livestock husbandry. This certainty is not yet available (Drewry *et al*. 2008). However, in this experiment it was shown that total macroporosity under HI–SD grazing management was maintained throughout the experiment and a continuous decline in total macroporosity occurred under SS management. This indicates that the rest periods used under this management regime, for this soil type and under the weather conditions that occurred, may have been appropriate.

There were not always significant differences in macropore properties between plots of treatments SS and HI–SD. This may be attributed to several factors. In particular, even set-stocked grazing can have a relatively benign impact on soil structural condition and pasture performance if weather conditions are favourable. During this experiment, the pastures were not subject to severe grazing, nor to extended dry or wet conditions. Presumably, if the weather during the experimental period had been distinctly wetter or drier, or the experiment had run for longer, the differences in topsoil structural condition under the different grazing treatments might have been more pronounced.

Although the soil structural data indicate that grazing tactics lead to subtle but discernible changes in topsoil physical condition, the practical significance of all of these changes is not clear. The greater total macroporosity observed under the HI–SD grazing regime and the pasture cages (treatment CA) means that water intake and storage will be greater than that under the SS grazing regime, but because the differences in macroporosity are most evident at the soil surface, the percolation of water into the subsoil may not differ greatly between treatment soils. The significant differences in macropore surface area between treatment soils by Year 3, evident in Fig. 3, may be a useful indicator of the total microbial habitat space, which Shestak and Busse (2005) define as comprising those pores 0.2–30 μm in diameter. Although the image analysis method used here could only discern macropores >65 μm, it is probable that a soil with a large value of macropore surface area will have a greater likelihood of containing many small pores. What constitutes an ‘improvement’ in habitable pore space for microbes, however, is not known; Shestak and Busse (2005) conclude that biological indices of soil quality are currently poorly linked to soil structural properties.

However, this investigation has confirmed that the continuous presence of livestock, under set-stocked management, is likely to alter topsoil structural attributes not only at the surface, but to at least 100 mm below the surface. The significantly lower total macroporosity at 100 mm depth in the SS soil compared with the HI–SD and CA treatment soils after 3 years (Table 4) suggests that the compaction effect may extend beyond that depth, even under this relatively moderate stocking rate. HI–SD rotational grazing management largely maintained soil structural condition over the course of this experiment, with the 3-year duration of the experiment of some significance. Greenwood *et al*. (1997) argued that the impacts of grazing are cumulative, and all grazing tactics are likely to have similar impacts in the long-term; however, their study focused on differences in stocking rate where all treatments were set-stocked, and did not consider rotational grazing management strategies. Many studies of soil–pasture–livestock interactions are of short duration, often only a single grazing event, and do not capture the potential benefits of grazing rest from rotational grazing management.

Conclusion

During a 3-year comparison of a set-stocked grazing regime, a high density-short duration grazing regime, an ungrazed control, and a caged pasture treatment, there were few easily observable, significant differences in topsoil physical condition. With only a few exceptions, values of indicators of topsoil physical condition such as bulk density, unsaturated hydraulic conductivity, and organic carbon content were similar across all treatments for all years. However, by examining topsoil structure of the grazing treatment soils using image analysis, several significant differences emerged, indicating some subtle effects of the grazing tactics. Soil of treatment HI–SD maintained a stable set of structural properties over the course of the experiment, whereas soil under treatment SS showed a decline in soil structural quality. ‘Best’ soil structure, defined here as that encompassing large values of macroporosity and macropore surface area, and exhibiting a range of pore sizes, was observed under treatment CA where regular defoliation of pasture was combined with an absence of livestock hoof pressure. The mechanisms contributing to these differences are likely to be a combination of consolidation, aggregate disruption, and repacking at the soil surface, caused by the presence, absence, and duration of livestock hoof pressure, and macropore construction by flora and fauna. It was not determined if these changes were sufficient to alter soil function (e.g. soil water retention/transmission, pasture root depth or density) and therefore pasture productivity. However, the declining trend in macroporosity under set-stocked grazing management indicates that soil function may be compromised at some time on this site and that grazing tactics are a contributing factor. It also appears likely that the presence of actively growing pasture may impart greater benefits to soil structure than ‘locking up’ pasture land (as simulated by treatment C). An understanding of the nature of macropores by way of image analysis, which was important in detecting differences in soil structural condition between treatments in this experiment, has the potential to provide useful information about the functionality of soil in conjunction with other soil physical, chemical, and biological data.

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